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W7-X and the sawtooth instability: towards realistic simulations of current-driven magnetic reconnection

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Abstract

Magnetic reconnection in W7-X is studied by means of global numerical simulations in a series of models of increasing complexity. The magnetic geometry ranges from that of a cylinder to the full three-dimensional field of W7-X, and the equations solved range from ideal magnetohydrodynamics (MHD) to gyrokinetics. We simulate plasmas from the first operation phase with electron cyclotron current drive (ECCD). These are characterized by an equilibrium magnetic field featuring an ECCD-distorted 'humped' profile of the rotational transform ι, with $\iota = 1$ in two radial locations. Such plasmas generally show sawtooth activity, hence motivating the present study. We pay particular attention to the role of equilibrium current density gradients in the destabilization of reconnecting modes. When the equilibrium temperature and density gradients are artificially suppressed (to eliminate the pressure gradient drive), the perturbed electrostatic potential is radially localized between the locations at which $\iota = 1$. This is shown with a purely collisionless gyrokinetic model, in cylindrical geometry. In the real toroidal geometry of W7-X, for a non-ideal MHD model including a uniform resistivity, electron inertia and (numerical) viscosity, the same qualitative behaviour is observed. In particular, even if a resonant $(m, n) = (1, -1)$ perturbation is initialized, the most unstable mode is the $(m, n) = (-4, 4)$, where m and n are the poloidal and toroidal mode numbers, respectively. Other modes are destabilized due to geometric coupling. The growth rate of this instability scales as $\eta^{1/3}$, where η is the plasma resistivity, thus suggesting that ECCD drives ideal MHD stable W7-X plasmas towards non-ideal marginality. An ideal magnetohydrodynamic analysis confirms the result. A fluid-kinetic hybrid version of the EUTERPE code shows that gyrokinetic ions have a stabilizing effect on these modes. For W7-X relevant collisionalities, the growth rate scales linearly with the electron skin depth, d_e. Implications of our results for sawtoothing W7-X operation are discussed.

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Keywords: magnetic reconnection, stellarator, sawtooth instability

(Some figures may appear in colour only in the online journal)

1. Introduction

Wendelstein 7-X [1–3] is a superconducting optimized stellarator [4], and, as such, not only enjoys low levels of neoclassical transport [5], but is also expected to feature favourable ideal magnetohydrodynamics (MHD) stability properties. Transport studies have been undertaken to quantify the quality of energy confinement [6–9], but enough heating power is not yet available to access the pressure (β) limit in most magnetic configurations, which is of the order of $\langle\beta\rangle \sim 5\%$ [10].

W7-X is a device conceptually rooted in certain key computational and theoretical ideas [11], in particular quasi-isodynamicity and optimization of MHD equilibrium and stability, but the experimental programme sometimes poses additional operational complications. For instance, periodic and repetitive crashes of the central electron temperature, similar to sawtooth crashes in tokamaks, emerged during divertor strikeline control experiments with electron cyclotron current drive (ECCD) [12]. Measurements from soft x-ray tomography and electron cyclotron emission showed that the crashes are preceded by weak oscillating precursors and a displacement of the plasma core consistent with an $(m, n) = (1, -1)$ mode, with m and n being the dominant poloidal and toroidal Fourier harmonics. This displacement occurs within 100 μs, followed by expulsion and redistribution of the core into the external part of the plasma. Furthermore, two types of crashes, with different frequencies and amplitudes were detected.

The emergence of sawteeth [13–16] in W7-X has therefore generated new interest in magnetic reconnection in stellarators. Strumberger and Günter have presented comprehensive linear stability studies of resistive modes for W7-X-type stellarator equilibria with ECCD [17]. Nonlinear simulations of W7-X-like ι-profiles in a cylinder have also been performed with a two-fluid model [18]. The same type of ι-profiles have been investigated to study the possibility of kinetic destabilization of infernal modes [19] (thus renewing interest also for kinetic studies of q-profiles with reversed shear in tokamaks [20]). Nikulsin *et al* [21] have proposed an extension of the JOREK code [22] to stellarator geometry, in order address nonlinear magnetic reconnection physics in W7-X, and the non-ideal reduced MHD equations for stellarator geometry have also been studied analytically [23].

W7-X was optimized for ideal MHD stability, but the question of non-ideal stability can also be important, in the light of the experimental observations mentioned above. Resistive and two-fluid effects [24–26] affect the linear stability threshold and are sometimes able to predict nonlinear saturated magnetic islands correctly [27, 28], but are not sufficient to capture certain kinetic phenomena like collisionless heating [28, 29] or the collisionless damping of the shear Alfvén continuum [30]. W7-X plasmas are hot enough [19] that the resistive reconnection scale, $\delta_\eta \approx 5 \times 10^{-3}$ cm, is much smaller than the ion Larmor radius $\rho_\mathrm{i} \approx 0.2$ cm and the electron skin depth $d_\mathrm{e} \approx 0.2$ cm. A realistic description of magnetic reconnection in W7-X should therefore be kinetic.

A key ingredient in magnetic reconnection studies is played by the profile of the rotational transform ι, which indicates how many poloidal turns a field line makes during each toroidal turn around the flux surface. Reconnection occurs when ι is rational, and the current density gradient then provides one of the two drives for reconnecting instabilities (the other being the pressure gradient). Knowledge of the actual experimental ι-profile is difficult to obtain, particularly in a stellarator, since it is a global quantity that depends on the field-line geometry throughout the entire plasma volume. Fortunately, however, numerical transport modelling of the toroidal current evolution resulting from ECCD and the bootstrap current can provide some guidance. Such calculations suggest that ECCD can result in a profile where two resonant flux surfaces with $\iota = 1$ appear within the plasma, see e.g. the equivalent of our figure 3 in reference [12].

In the development of sawtooth dynamics in W7-X, which is sudden and characterized by two different types of crashes [12], the ι-evolution seems to regulate the access to different, but co-existing, unstable scenarios. In this article, we focus on the effect of the equilibrium current density gradients for a resonant ι-profile. The study of pressure-gradient-driven modes (not necessarily restricted to low mode numbers), is left for a future companion article, where kinetic effects will be investigated further.

In this work, by using the global codes GYGLES [31, 32] and EUTERPE [33], we simulate typical plasmas from the first operation phase in W7-X with ECCD. Our simulations feature several levels of complexity, both concerning physics and geometry. In addition to ideal MHD computations, we have used a purely collisionless gyrokinetic model in cylindrical geometry, and a non-ideal fluid model including uniform resistivity, electron inertia, (numerical) viscosity and gyrokinetic ions, in real W7-X geometry.

With two kinetic species, and in cylindrical geometry, it is found that the instability occurs after collisionless damping of the shear Alfvén continuum takes place. The radial positions at which $\iota = 1$ play an important role in the mode localization, and for the strength of the instability. Qualitatively the same localization are observed in cylindrical and stellarator geometry. With the non-ideal fluid model, in W7-X, long-wavelength reconnecting instabilities for an ECCD-distorted ι-profile show growth rates that follow the scaling [34] $\eta^{1/3}$, where η is the plasma resistivity, thus suggesting that ECCD drives ideally MHD stable W7-X plasmas towards non-ideal marginality. This hypothesis is confirmed by an ideal MHD study performed with the stability code CAS3D [35], which indeed predicts ideal stability in W7-X. Moreover, the ideal

mode closest to marginality evaluated with CAS3D is the dominant non-ideal instability evaluated by the code EUTERPE. A fluid-kinetic hybrid version of the EUTERPE code shows that gyrokinetic ions have a stabilizing influence on these modes, but do not suppress them. For W7-X relevant collisionalities, the mode transitions to a collisionless regime with a growth rate scaling linearly with the electron skin depth, d_e, reproducing the typical low-β cylindrical result [23].

The article is structured as follows. Ideal MHD stability properties of the plasma equilibria, subsequently studied within the gyrokinetic picture, are discussed in section 2. Kinetic equations and numerical methods are presented in section 3. Numerical results are given in section 4. Here, we first discuss the effect of the 'humped' rotational transform in a cylindrical plasma with two kinetic species. We then present our magnetic reconnection study for the true W7-X geometry. Conclusions are summarized in the final section.

2. Preliminaries: ideal MHD stability for W7-X-like ECCD ι-profiles

Before presenting our gyrokinetic and nonideal analysis, we must ask ourselves what does ideal MHD say about W7-X-like ι-profiles modified by ECCD, and whether that is enough to understand the real W7-X stability properties under sawtoothing conditions. Let us therefore consider the stability of two types of plasma equilibria by linear ideal mode analysis based on the ideal MHD energy principle [36, 37]. Magnetic equilibria are evaluated with the VMEC equilibrium code [38, 39], whereas the stability analysis is performed with the eigenvalue code CAS3D [35]. Both codes are used in their free-boundary versions. A detailed description of the combined equilibrium/stability analysis procedure is given in reference [40].

Firstly, the stability properties of an internal kink are studied in a screw-pinch equilibrium with all basic parameters chosen similar to W7-X values. While it is known that toroidal geometry has a critical impact on the stability of the linear ideal MHD kink mode [41], the cylindrical limit is still important if one aims at a non-ideal understanding of the sawtooth phenomenon in W7-X, let alone to disentangle the small scales present in the system [23]. The full equilibrium description is given in appendix A, where table 2 summarizes the parameters of the rotational transform profile in (A1) including the ι-hump modelling ECCD. Notice that in all the MHD simulations the pressure gradient is not artificially suppressed. For the screw pinch studied here, the rotational transform is given by

$$\iota(r) = 0.88 + 0.115(r/a)^2 + f_{\rm exp}\ \exp[-(r/a - s_c)^2/\sigma], \quad (1)$$

with r/a the normalized minor radius and s_c the center of the hump. The resonance condition in straight-field-line coordinates, $k_\parallel = m\,\iota + n = 0$, shows that perturbations with poloidal and toroidal Fourier harmonics, $m = -n$, are resonant at the two locations where $\iota = 1$. Due to the cylindrical geometry and the linear nature of the equations, all perturbation harmonics may be studied separately in the stability calculation. Here, the focus is on $m = 1$ modes. Variations of the rotational transform are studied by changing (i) the radial location s_c and (ii) the height of the of the ι-hump controlled by $f_{\rm exp}$. The parameter σ is adjusted to have the locations of $\iota = 1$ at a distance of 0.2 in the normalized minor radius. The various settings of these ι-profile parameters are summarized in table 2. The corresponding profiles are shown in the left frame of figure 1, with the ι-hump centers at the radial locations $s_c \approx 0.3, 0.4$, and 0.5, fixing $f_{\rm exp}$ to 0.2. By way of example, the normal displacement for the latter case, $s_c \approx 0.5$, is shown, too, with clear signatures of an internal kink mode being confined between the resonances and having steep gradients at these locations. The convergence of the instability growth rate with the radial grid size is shown in the right frame of figure 1. The closer the ι-hump comes to the magnetic axis, the smaller the growth rate becomes. In the cases considered here it varies between $1.4\,(\mathrm{ms})^{-1}$ for $s_c \approx 0.5$ and $0.86\,(\mathrm{ms})^{-1}$ for $s_c = 0.3$. Obviously, reducing the height of the ι-hump also has a stabilising effect. With $s_c \approx 0.5$ the growth rate decreases to $0.95\,(\mathrm{ms})^{-1}$ when changing $f_{\rm exp}$ from 0.2 to 0.12, a reference value of the gyrokinetic calculation of section 4.1. Transforming the ideal-MHD growth rate to the units used in the kinetic case, yields $950\,\mathrm{s}^{-1}$. As we shall see, this compares well to the gyrokinetic result, the ideal MHD result being more unstable.

Secondly, with the parameter set-up described in appendix A, a finite-plasma-β W7-X-type equilibrium is considered for its ideal MHD properties. In section 4.2, this equilibrium will again be studied using a hybrid fluid-kinetic model. Three characteristic plasma cross-sections of the equilibrium are shown in figure 7, at the beginning, at a quarter, and half of a field period of the five-periodic plasma domain.

Unlike the screw-pinch, stellarator fields possess rotational transform already without a net toroidal current (black dotted line in figure 2). This rotational transform profile is modified with an ι-hump, which is here centered at the radial location $s_c \approx 0.4$ and with $\iota_{\rm max} = 1.02$, to again model ECCD (solid black line in figure 2). The result of a free-boundary calculation in which the plasma boundary is allowed to move is shown in figure 2. Each of the three scalar displacement components is approximated employing 166 Fourier harmonics belonging to the so-called $N = 1$ mode family [10], with the poloidal Fourier harmonics $0 \leqslant m \leqslant 19$, the toroidal Fourier harmonics $-16 \leqslant n \leqslant 11$, and 500 flux intervals. The energy in the perturbed magnetic field is used as norm for the eigenmodes, facilitating the determination of stability limits without the interference of the stable ideal MHD continua [42]. The equilibrium is found to be stable with the chosen parameter setting concerning flux- and Fourier resolution and mode family of the perturbations. Figure 2 shows the ten largest Fourier harmonics of the mode closest to marginal stability, among them the $m = -n = 6, 4, 9$, and 11 amplitudes are strongest. Stellarator-type helical equilibrium terms couple the $m = -n = 6$ and $m = -n = 11$ harmonics and, likewise, the $m = -n = 4$ and $m = -n = 9$ harmonics. Comparing the $m = -n = 6$ and $m = -n = 11$ amplitudes, the latter is smaller only because of the finite size of the perturbation Fourier table. In figure 2, the dominant harmonics show clear signs of an internal kink with only small side-bands. The most important of the side-bands with amplitudes about a factor of ten smaller than

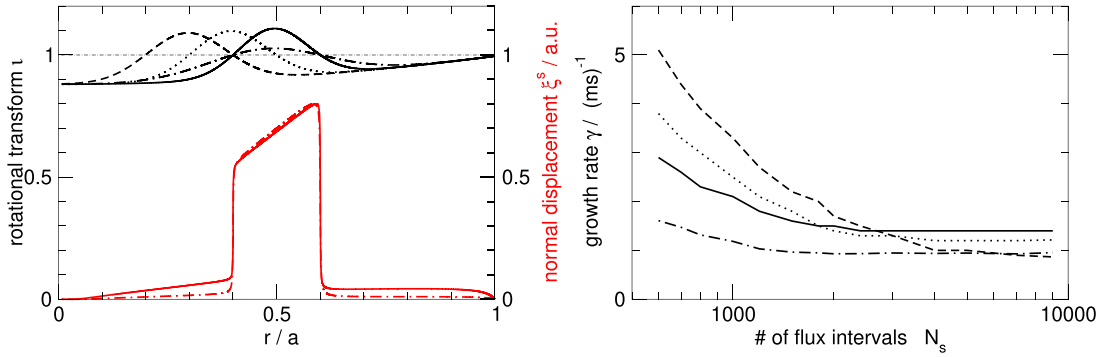


Figure 1. Ideal MHD results for a straight, cylindrical plasma with minor radius $a = 0.55m$, aspect ratio $A = L/a = 2\pi \times 10$, $B_0 = 2.5$ T (compare appendix A). (Left) Rotational transform profiles (black), each with two intersections of $\iota = 1$, and normal displacements (red), $\boldsymbol{\xi} \cdot \nabla s$, with $m = 1$, $n = -1$ for the $s_c = 0.5$ profiles. (Right) Dependence of the growth rate in units of $(\text{ms})^{-1}$ on the number of flux intervals. Legend for both sub-frames: $s_c = 0.5$ and $f_{\text{exp}} = 0.2$ (solid); $s_c = 0.4$ and $f_{\text{exp}} = 0.2$ (dotted); $s_c = 0.3$ and $f_{\text{exp}} = 0.2$ (dashed); $s_c = 0.5$ and $f_{\text{exp}} = 0.12$ (dash-dotted). Code used: CAS3D. Model: ideal MHD, cylindrical geometry.

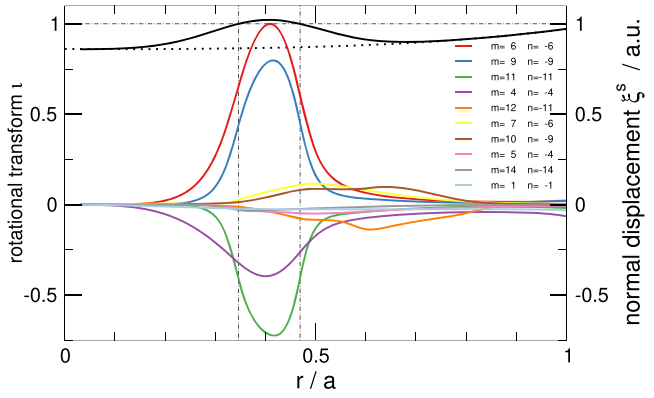


Figure 2. Ideal MHD result for a W7-X plasma: Normal displacement harmonics of a stable, free-boundary mode versus normalized effective minor radius. Smaller side-bands are indicated too. Rotational transform profiles are also shown (black): without (dotted) and with (solid) ECCD-model contribution. Colour legend: $\xi^s_{6,-6}$ (red), $\xi^s_{4,-4}$ (dark violet), $\xi^s_{9,-9}$ (blue), $\xi^s_{11,-11}$ (green). Computation parameters: 500 flux intervals, 166 perturbation harmonics, ι-hump with $s_c \approx 0.4$, $f_{\text{exp}} \approx 0.15$, and $\sigma \approx 0.024$, see table 2. Code used: CAS3D. Model: ideal MHD, W7-X geometry.

the maximum represent tokamak-type couplings invoked by toroidicity, e.g. $(m, n) = (7, -6)$ coupled to $(m, n) = (6, -6)$ or $(m, n) = (10, -9)$ to $(m, n) = (9, -9)$. As indicated by the small amplitudes of the normal displacement harmonics near the plasma boundary, $r/a = 1$, this mode does not exploit the boundary condition but is essentially a fixed-boundary mode. With the ι-hump relatively far from the plasma boundary, this is understandable and is confirmed by the nearly identical result from the corresponding fixed-boundary stability analysis, in which the ideal MHD displacement leaves the plasma boundary unchanged.

We conclude that, while the cylindrical ideal case (where the rotational transform is entirely generated by plasma current) is useful to infer unstable mode structures, its ideal MHD stability properties are different from those for a similar ι-profile in true W7-X geometry, where most of the rotational transform is produced by external currents and only stable ideal modes are found.

3. Kinetics: equations and methods

Having seen that an ECCD-distorted ι-profile is not enough to destabilise W7-X in ideal MHD, we now proceed to a gyrokinetic description of the plasma.

The standard formulation of the gyrokinetic equations for current-driven modes in the Hamiltonian $p_\parallel$-formulation has been described in detail in [30, 43]. For derivation of the fluid equations, considered below, it is more convenient to use the $v_\parallel$-formulation of the gyrokinetic theory. To assess the current driven modes, we choose the equilibrium electron distribution function to be a Maxwellian that is shifted in velocity space by an amount corresponding to the parallel current:

$$F_{0\text{e}} = n_0 \left(\frac{m_\text{e}}{2\pi T_\text{e}}\right)^{3/2} \exp\left[-\frac{m_\text{e}(v_\parallel - u)^2}{2T_\text{e}}\right] \times \exp\left[-\frac{m_\text{e} v_\perp^2}{2T_\text{e}}\right]. \tag{2}$$

Here $u = -j^{(0)}_\parallel/(en_0)$ and $\mu_0 j^{(0)}_\parallel = (\mathbf{B}/B)\cdot(\nabla \times \mathbf{B})$ with $\mathbf{B}$ the equilibrium magnetic field. The ion equilibrium distribution is assumed to be a stationary Maxwellian, so that all equilibrium current is carried by the electrons.

The perturbed distribution function δf_s of each species s is found from the gyrokinetic equation:

$$\frac{\partial \delta f_s}{\partial t} + \dot{\mathbf{R}} \cdot \frac{\partial \delta f_s}{\partial \mathbf{R}} + \dot{v}_\parallel \frac{\partial \delta f_s}{\partial v_\parallel} = -\delta\dot{\mathbf{R}} \cdot \frac{\partial F_{0s}}{\partial \mathbf{R}} - \delta\dot{v}_\parallel \frac{\partial F_{0s}}{\partial v_\parallel}, \tag{3}$$

where $\delta f_s \ll F_{0s}$, with F_{0s} the equilibrium distribution function, and $v_\parallel = \hat{\mathbf{b}} \cdot \mathbf{v}$. Here $\hat{\mathbf{b}} = \mathbf{B}/B$ and $\mathbf{v}$ is the particle velocity. The coefficients in equation (3) are determined by the equations of motion of gyrocentres

$$\begin{aligned}\dot{\mathbf{R}} &= v_\parallel \hat{\mathbf{b}} + \frac{1}{q_s}\left[\frac{\mu}{B^*}\hat{\mathbf{b}} \times \nabla B + \frac{m_s v_\parallel^2}{B^*}\hat{\mathbf{b}} \times \boldsymbol{\kappa}\right] \\ &\quad + \frac{v_\parallel}{B^*}\hat{\mathbf{b}} \times \boldsymbol{\kappa}\langle \delta A_\parallel \rangle + \frac{1}{B^*}\hat{\mathbf{b}} \times \nabla \langle \delta\phi - v_\parallel \delta A_\parallel \rangle \\ &\equiv \dot{\mathbf{R}}^{(0)} + \delta\dot{\mathbf{R}},\end{aligned} \tag{4}$$

and

$$m_s\dot{v}_\parallel = -\mu\nabla B\cdot\left[\hat{\boldsymbol{b}}+\frac{m_s}{q_s}\frac{v_\parallel}{B^*}\hat{\boldsymbol{b}}\times\boldsymbol{\kappa}\right] - q_s\left[\hat{\mathbf{b}}\cdot\nabla\langle\delta\phi\rangle+\frac{\partial}{\partial t}\langle\delta A_\parallel\rangle\right] - \frac{m_s v_\parallel}{B^*}\hat{\boldsymbol{b}}\times\boldsymbol{\kappa}\cdot\nabla\langle\delta\phi\rangle - \frac{1}{B^*}\mu\nabla B\cdot\left[\nabla\langle A_\parallel\rangle\times\hat{\boldsymbol{b}}+\langle A_\parallel\rangle\hat{\boldsymbol{b}}\times\boldsymbol{\kappa}\right] \equiv -\mu\nabla B\cdot\left[\hat{\boldsymbol{b}}+\frac{m_s}{q_s}\frac{v_\parallel}{B^*}\hat{\boldsymbol{b}}\times\boldsymbol{\kappa}\right]+m_s\delta\dot{v}_\parallel, \tag{5}$$

where m_s is the particle mass, $\mu = m_s v_\perp^2/(2B^2)$ the magnetic moment, with $v_\perp^2 = v^2 - v_\parallel^2$, and $\delta\phi$ and $\delta A_\parallel$ are the perturbed electrostatic and parallel magnetic potentials. Here $\langle\cdot\rangle$ is the gyroaverage, B^* is given by $B^* = B + \left(\frac{m_s}{q_s}v_\parallel + \delta A_\parallel\right)\hat{\boldsymbol{b}}\cdot(\nabla\times\hat{\boldsymbol{b}})$ and the curvature is $\kappa = \widehat{\boldsymbol{b}}\cdot\nabla\widehat{\boldsymbol{b}}$. Physically, gyrocentres move along the total magnetic field with the velocity $v_\parallel$ under the influence of the magnetic mirror force and the perturbed electric field in equation (5). Across the field, the movement is given by the sum of grad-B, curvature, and perturbed $\mathbf{E}\times\mathbf{B}$ drifts. The system of equations is closed by the quasineutrality equation and Ampère's law:

$$\int\frac{q_i^2 F_{0i}}{T_i}(\delta\phi-\langle\delta\phi\rangle)\delta(\mathbf{R}+\boldsymbol{\rho}-\mathbf{x})\mathrm{d}^6Z = \sum_s q_s n_{1s}, \tag{6}$$

$$-\nabla_\perp^2\delta A_\parallel = \mu_0\sum_s j_{\parallel 1s}, \tag{7}$$

where $n_{1s} = \int \mathrm{d}^6Z\,\delta f_s\delta(\mathbf{R}+\boldsymbol{\rho}-\mathbf{x})$ is the gyrocentre density and

$$j_{\parallel 1s} = q_s\int \mathrm{d}^6Z\delta f_s v_\parallel\delta(\mathbf{R}+\boldsymbol{\rho}-\mathbf{x})$$

the gyrocentre current. Here, q_s is the electric charge, $\mathrm{d}^6Z = B_\parallel^*\,\mathrm{d}\mathbf{R}\,\mathrm{d}v_\parallel\mathrm{d}\mu\,\mathrm{d}\theta$ is the phase-space volume element and $\boldsymbol{\rho}$ is the gyroradius vector.

Note that the polarization density in equation (6) is given by an integral operator and includes a non-local effect of the ion gyro-orbit. In this work, we use the long-wavelength approximation of the polarisation density.

For gyrokinetic simulations, we employ the global code GYGLES [31, 32], which features the mixed-variable formulation [44, 45], that is, the vector potential is expressed as a sum $\delta A_\parallel = \delta A_\parallel^{(\mathrm{h})} + \delta A_\parallel^{(\mathrm{s})}$, where the time derivative of the 'symplectic part' $\delta A_\parallel^{(\mathrm{s})}$ is taken to be given by the ideal Ohm's law [44, 45] of the magnetic potential:

$$\frac{\partial}{\partial t}\delta A_\parallel^{(\mathrm{s})} + \hat{\mathbf{b}}\cdot\nabla\delta\phi = 0. \tag{8}$$

Furthermore, a mixed velocity is introduced as $u_\parallel = v_\parallel + (e_s/m_s)\left\langle\delta A_\parallel^{(\mathrm{h})}\right\rangle$. The perturbed part of the distribution function is then discretised by a total of N_p markers:

$$\delta f_s(\mathbf{R},u_\parallel,\mu,t) = \sum_{\nu=1}^{N_\mathrm{p}} w_{s\nu}(t)\delta(\mathbf{R}-\mathbf{R}_\nu)\delta(u_\parallel-p_{\parallel\nu})\delta(\mu-\mu_\nu), \tag{9}$$

where $(\mathbf{R}_\nu, u_{\parallel\nu}, \mu_\nu)$ are the marker phase-space coordinates and $w_{s\nu}$ is their weight. The electrostatic and magnetic potentials are discretised with the finite-element method (Ritz–Galerkin scheme):

$$\delta\phi(\mathbf{x}) = \sum_{l=1}^{N_s}\delta\phi_l\Lambda_l(\mathbf{x}) \qquad \delta A_\parallel(\mathbf{x}) = \sum_{l=1}^{N_s} a_l\Lambda_l(\mathbf{x}), \tag{10}$$

where $\Lambda_l(\mathbf{x})$ are finite elements (tensor products of B splines [46, 47]), N_s is the total number of the finite elements, and $\delta\phi_l$ and a_l are the spline coefficients. Homogeneous Dirichlet boundary conditions are applied for $\delta\phi$ and $\delta A_\parallel$ at the plasma edge. For further numerical details, see references [31, 32, 48–51].

All our gyrokinetic calculations are performed in a screw-pinch geometry for two kinetic species: ions and electrons. For W7-X, however, the complexity of the geometry, and the nature of the reconnection instabilities that we study (global in nature and yet mediated by small physical scales), compel us to resort to a hybrid version of the kinetic code EUTERPE [33], whose equations are derived in the following way.

We follow Cole *et al* [52], and form the first two fluid moments of the gyrokinetic equation. By taking the velocity-space integral of equation (3) for electrons, one obtains the electron continuity equation [52]

$$\frac{\partial\delta n_\mathrm{e}}{\partial t} + n_0\mathbf{B}\cdot\nabla\left(\frac{u_{\parallel\mathrm{e}}}{B}\right) = \frac{n_0}{B}\hat{\mathbf{b}}\times\nabla\left(\delta\phi-\frac{2}{n_0 e}\delta p_\mathrm{e}\right)\cdot\frac{\nabla B}{B} + \frac{\nabla\times\mathbf{B}}{B^2}\cdot\left(n_0\nabla\delta\phi-\frac{1}{e}\nabla\delta p_\mathrm{e}\right) - B\mathbf{v}_\mathrm{E}\cdot\nabla\left(\frac{n_0}{B}\right) - \left(\nabla\times\delta A_\parallel\hat{\mathbf{b}}\right)\cdot\nabla\left(\frac{n_0}{B}u_{\parallel\mathrm{e}}^{(0)}\right), \tag{11}$$

where $\mathbf{v}_\mathrm{E} = B^{-1}\hat{\mathbf{b}}\times\nabla\delta\phi$, when finite electron Larmor radius effects are neglected and an isotropic pressure tensor with $\delta p_\mathrm{e} = n^{-1}\int \mathrm{d}^3\mathbf{v}m_\mathrm{e}v_\parallel^2\delta f_\mathrm{e}$ is assumed. By evaluating the parallel velocity moment of the electrons, one obtains the generalised Ohms's law

$$\frac{\partial}{\partial t}\left(\delta A_\parallel - d_\mathrm{e}^2\nabla^2\delta A_\parallel\right) = -\widehat{\mathbf{b}}\cdot\nabla\delta\phi - \frac{1}{q_\mathrm{e}n}\widehat{\mathbf{b}}\cdot\nabla\delta p_\mathrm{e} + \frac{\widehat{\mathbf{b}}\cdot\nabla B}{enB}\times\int\mathrm{d}^3\mathbf{v}v_\parallel\mu\delta f_\mathrm{e} - \frac{T_\mathrm{e}}{q_\mathrm{e}B}\frac{\widehat{\mathbf{b}}\times\nabla p_\mathrm{e}}{p_\mathrm{e}}\cdot\nabla\delta A_\parallel, \tag{12}$$

where $p_\mathrm{e} = nT_\mathrm{e}$ and $d_\mathrm{e}^2 = m_\mathrm{e}/(\mu_0 n e^2)$ denotes the electron skin depth. For homogeneous magnetic equilibria, equations (11) and (12) are the fluid components of the linear limit of the KREHM model [28, 29] with equilibrium background gradients [53, 54]. The KREHM model treats kinetic electrons in a fluid fashion, by allowing higher order moments of the

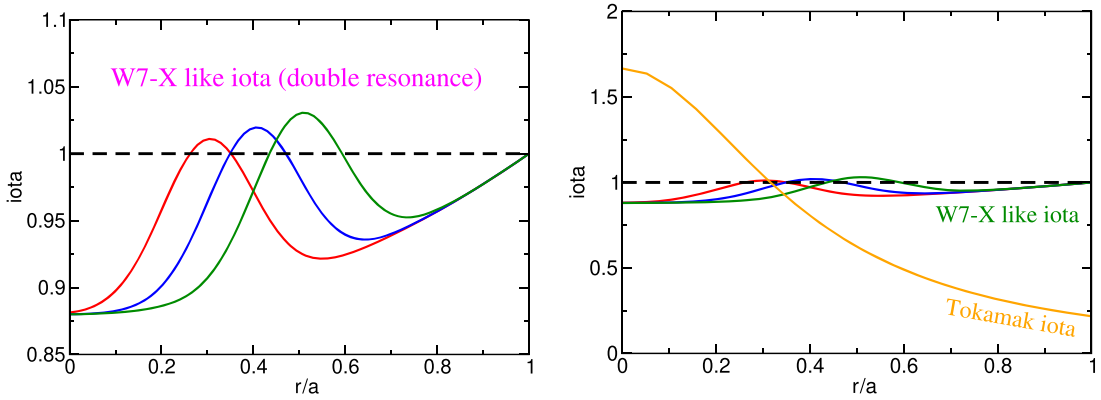


Figure 3. (Left) A set of W7-X like rotational transform with two resonant flux surfaces modelling the late-time ι behaviour with ECCD. (Right) Comparison of the low-shear W7-X like rotational transform profiles to a typical sawtoothing tokamak case.

perturbed distribution function to evolve following a recursive equation, their effect being encapsulated in the pressure term δp_e. The system of equations is closed by using quasineutrality with the ion density $\delta n_\mathrm{i} = \int \mathrm{d}^3\mathbf{v}\delta f_\mathrm{i}$ and the perturbed ion distribution function δf_i. Practically, one only needs to replace δn_i in the electron continuity equation just evaluated. Such a model, with isothermal electrons, $\delta T_\perp = \delta T_{\parallel\mathrm{e}} = \int \mathrm{d}^3\mathbf{v} v_\parallel^2 \delta f_\mathrm{e} \equiv 0$, has been implemented and will be the reference hybrid model for future kinetic studies with finite equilibrium pressure gradients. In this work, we retain the continuity equation for the electrons, (11), while the ions are treated either fully gyrokinetically or using a reduced non-ideal MHD model that is derived in the following way.

We evaluate the continuity equation for the ions, and subtract the result from the electron continuity equation, to obtain

$$\frac{\partial \delta n}{\partial t} + \nabla\cdot\left[n_0\hat{\mathbf{b}}u_\parallel + \frac{1}{eB}j_\parallel^{(0)}\hat{\mathbf{b}}\times\nabla\delta A_\parallel - \frac{1}{eB^2}j_\parallel^{(0)}\left(\hat{\mathbf{b}}\times\nabla B + (\nabla\times\mathbf{B})_\perp\right)\delta A_\parallel - \frac{2}{eB^2}\left(\hat{\mathbf{b}}\times\nabla B + (\nabla\times\mathbf{B})_\perp\right)\delta p\right] = 0, \quad (13)$$

with $\delta n = \delta n_\mathrm{e} - Z\delta n_\mathrm{i}$, $\delta p = \delta p_\mathrm{i} + \delta p_\mathrm{e}$, $u_\parallel = u_{\parallel\mathrm{e}} - Zu_{\parallel\mathrm{i}}$, and $\mu_0 j_\parallel^{(0)} = \hat{\mathbf{b}}\cdot\nabla\times\mathbf{B}$. Ohm's law is taken to be consistent with the isothermal limit of the true electron parallel momentum equation, equation (12),

$$\frac{\partial}{\partial t}\left(\delta A_\parallel - d_\mathrm{e}^2\nabla^2\delta A_\parallel\right) = -\hat{\mathbf{b}}\cdot\nabla\delta\phi + \frac{\eta}{\mu_0}\nabla_\perp^2\delta A_\parallel, \quad (14)$$

where we have added a resistive term proportional to the Spitzer resistivity, η, and we have specialized to flat pressure profiles. Notice that we also neglect the ion FLR correction $1/(q_\mathrm{e}n)\hat{\mathbf{b}}\cdot\nabla T_\mathrm{e}\delta n_\mathrm{e}$, for reasons that will become apparent. A relation between $u_\parallel$ and $A_\parallel$ is given by the parallel component of Ampére's law

$$u_\parallel = \frac{1}{\mu_0 n_0 e}\nabla\cdot\nabla_\perp\delta A_\parallel. \quad (15)$$

The electrostatic potential is determined by the quasineutrality condition

$$\nabla\cdot\left(\frac{m_\mathrm{i}n_0}{B^2}\nabla_\perp\delta\phi\right) = e\delta n, \quad (16)$$

and the system is closed using the flat-density pressure equation of reduced MHD [23]

$$\frac{\partial\delta p}{\partial t} = -\mathbf{v}_\mathrm{E}\cdot\nabla p_0 + \nu\nabla_\perp^2\delta p, \quad (17)$$

where a viscosity, ν, is added for reasons of numerical stability. Equations (13)–(17) form the fundamental model studied below.

A higher-fidelity model is obtained by solving the gyrokinetic equation for the ions and inserting their density perturbation on the right-hand side of equation (16),

$$\nabla\cdot\left(\frac{m_\mathrm{i}n_0}{B^2}\nabla_\perp\delta\phi\right) = e\delta n_\mathrm{e} - Ze\left\langle\delta n_\mathrm{i}\right\rangle, \quad (18)$$

and equation (15) with

$$u_{\parallel\mathrm{e}} = \frac{1}{en_0}\delta j_{\parallel\mathrm{i}} + \frac{1}{\mu_0 n_0 e}\nabla\cdot\nabla_\perp\delta A_\parallel, \quad (19)$$

where the quantities δn_i and $\delta j_{\parallel\mathrm{i}}$ are evaluated as velocity space integrals of the perturbed ion distribution function, obtained from equation (3). In this case, δn_e follows equation (11), with δp_e determined by [52]

$$\frac{\partial\delta p_\mathrm{e}}{\partial t} = -\mathbf{v}_\mathrm{E}\cdot\nabla p_\mathrm{e} + \nu\nabla_\perp^2\delta p_\mathrm{e}. \quad (20)$$

The hybrid kinetic model closed by equation (20) has proven to work well for nearly Alfvénic perturbations in W7-X, and for kink instabilities in cylindrical and tokamak geometry [52, 55]. It is here adopted on the basis of its success in these contexts. There is a small inconsistency in treating ions gyrokinetically and whilst neglecting the ion finite-Larmor-radius term in the parallel momentum equation for the electrons. (This term comes from the density perturbation, which is taken from the ions, in the perturbed electron pressure.) While this could seem an issue, in reality is a useful simplification which allows us to formally explore some regimes of interest for the instabilities that we are discussing. The correction to Ohm's law due to the

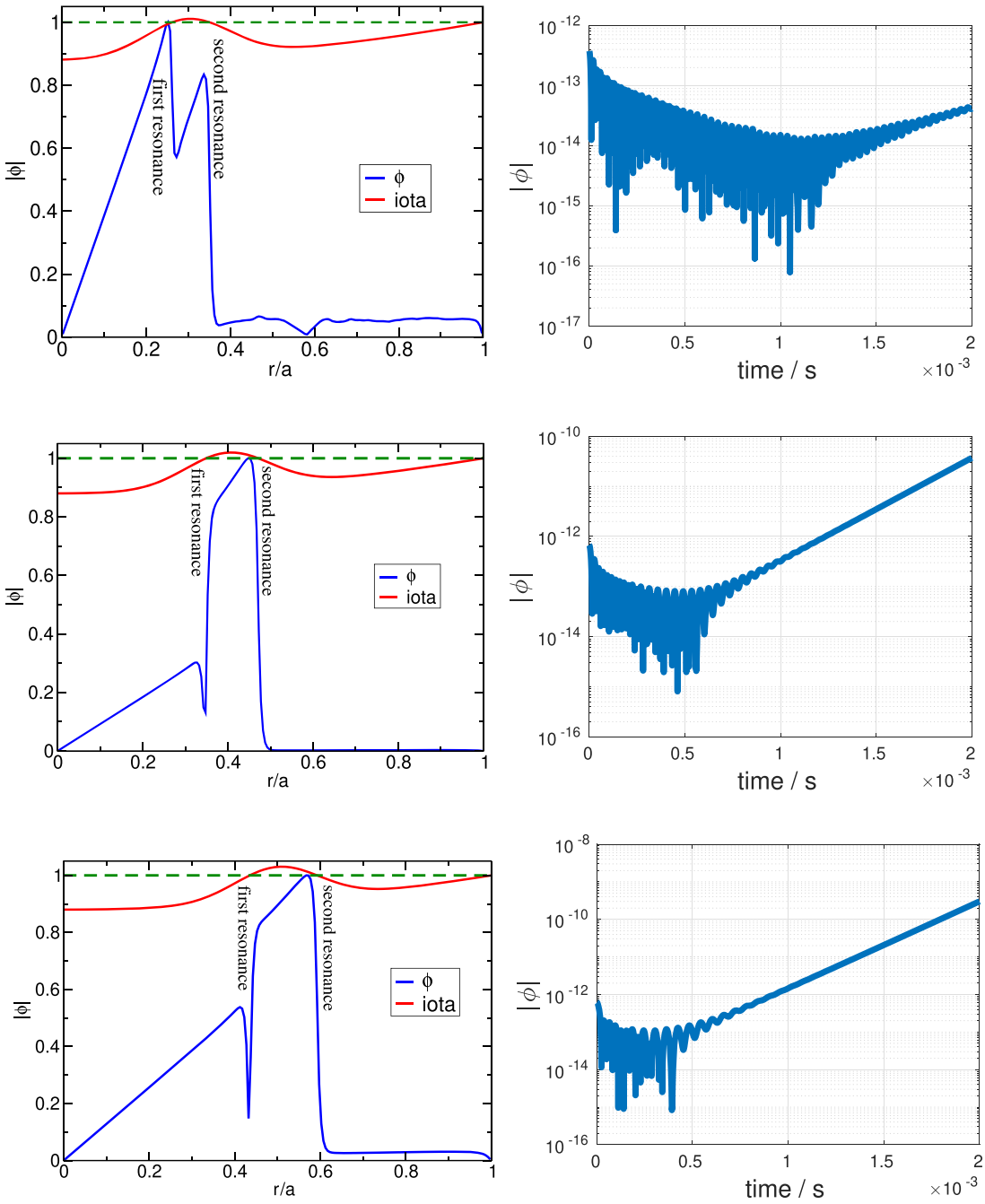


Figure 4. (Left) Mode structure of the 'double-kink' instability for the rotational transform shown (corresponding to $s_c = 0.3$, 0.4 and 0.5). Right: evolution of the instability. One can see how the shear Alfvén wave continuum decays and the unstable 'double-kink' mode emerges. The growth rate is larger when the resonant flux surfaces are shifted towards the plasma edge. Code used: GYGLES. Model: gyrokinetics, two species, cylindrical geometry.

full ion Larmor radius response is known, it can even be treated analytically [56, 57], and would only add some unnecessary subtleties which are not particularly interesting at this point. In particular, we would obtain a collisionless scaling for a growing $m = 1$ mode of the type $\gamma \propto d_e^{1/3}\rho_i^{2/3}$, rather than $\gamma \propto d_e$ [29, 58]. However, one new interesting aspect of ion kinetics in reconnecting modes theory has been recently pointed out by Connor *et al* [59], and is the effect of ion Landau damping. The ion kinetic model here used includes this physics.

We conclude this section by noting that the equations are solved with the Fourier version of the EUTERPE solver. This is based on a discrete Fourier transform accomplished through the projection of the perturbed equations over a finite set of B-splines for the radial, s, poloidal, θ, and toroidal, φ, variables.

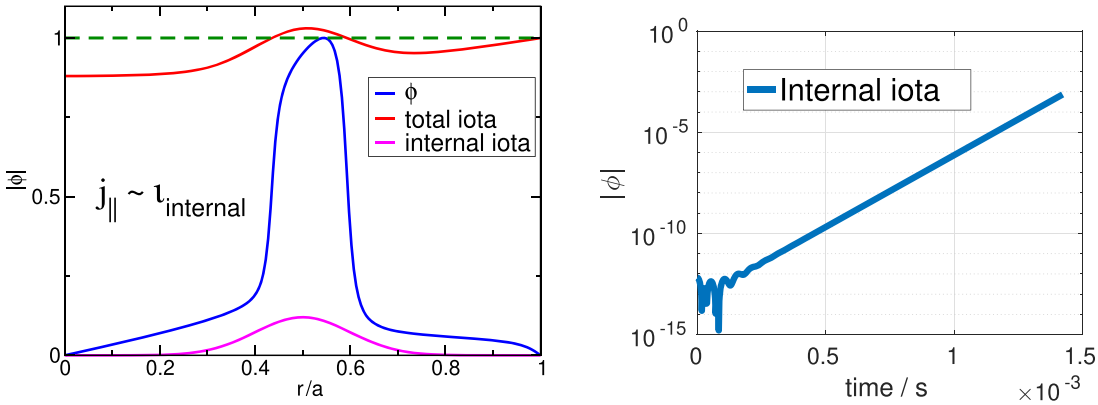


Figure 5. (Left) Mode structure of the 'double-kink' instability for the rotational transform shown. Only the internal $\iota_{\rm pl}$ is used for the ambient parallel current, see equation (27). Right: evolution of the instability. One can see how the shear Alfvén wave continuum decays and the unstable 'double-kink' mode emerges. The mode drive appears to be even stronger than in the previous case shown in figure 4. Code used: GYGLES. Model: gyrokinetics, two species, cylindrical geometry.

4. Results

4.1. Gyrokinetic treatment in cylindrical geometry

Let us now consider internal kink instabilities in a straight cylinder with low-shear rotational transform profiles that possess a local maximum, such as those shown in figure 3. Note that two resonant flux surfaces with $\iota = 1$ appear within the plasma. It is then reasonable to expect a destabilization of current-driven internal kink modes, just as in tokamaks.

As in W7-X, we take the minor radius to be $a = 0.55$ m, the major radius $R_0 = 5.5$ m, the magnetic field $B_0 = 2.5$ T, and the rotational transform ($\rho = r/a$)

$$\iota = 0.88 + 0.12\rho^2 + 0.12\,\exp\left[-\frac{(\rho - s_{\rm c})^2}{0.02}\right] \qquad (21)$$

The flux surface $s_{\rm c}$ corresponds to the maximum of the rotational transform profile distortion by the plasma current. We assume a flat density profile $n_0 = 2\times 10^{19}$ m^{-3}, flat ion and electron temperatures, $T_{\rm i} = 2$ keV and $T_{\rm e} = 5$ keV. The radial variation has been set to zero in order to isolate current-driven modes in these simulations, but otherwise similar plasma densities and temperatures have been measured in recent W7-X experimental campaigns [1].

In figure 4, we show the linear evolution of the electrostatic potential calculated by GYGLES [32] in the configuration considered. One sees how the most unstable mode grows after the decay of the shear Alfvén wave continuum. This mode is radially localized to the region where ι is resonant and its structure is that of the ideal mode of figure 1. This is somewhat obfuscated by the fact that here we are showing the electrostatic potential, whereas in figure 1 we show the contravariant component of the plasma displacement. We stress that one should not expect complete quantitative agreement between the ideal and the kinetic computations, if kinetic effects are not negligible [30]. The growth of the instability is stronger when the resonant flux surfaces are shifted towards the plasma edge, i.e. for $s_{\rm c} = 0.5$ in equation (21), and ι exceeds unity, as for the ideal-MHD case of section 2. The mode is driven by the gradient of the parallel plasma current density.

The usual relation between the equilibrium parallel plasma current and the rotational transform in screw pinch geometry follows from the equations

$$\mathbf{B} = \nabla\psi\times\nabla\varphi + F(\psi)\nabla\varphi,\ \psi(r) = \frac{B_0 r^2}{2},\ F(r) = \frac{B_0 r^2}{R_0}\iota(r) \qquad (22)$$

$$j_\parallel = \frac{1}{\mu_0 r}\frac{{\rm d}F}{{\rm d}r} = \frac{B_0}{\mu_0 R_0}\iota(2-\widehat{s}) = \frac{B_0}{\mu_0 R_0}(2\iota + r\iota'),\quad \widehat{s} = -\frac{r\iota'}{\iota}. \qquad (23)$$

Notice that our angles are kept consistent with a toroidal system, thus the traditional 'poloidal' angle of a screw pinch is here our φ.

This cylindrical model differs in an important way from W7-X, where a large part of ι is provided by currents in external coils rather than by the plasma current:

$$\iota = \iota_{\rm ext} + \iota_{\rm pl},\qquad \iota_{\rm ext} \gg \iota_{\rm pl}. \qquad (24)$$

One may therefore question whether the plasma current itself can provide enough rotational transform for instability to occur. The drive for the instability comes from the gradient in the current density, which contributes to the magnetic shear $\iota'_{\rm pl}(r)$, whereas the shear $\iota'_{\rm ext}(r)$ in the external transform does not drive the instability. The magnetic shear is mostly due to plasma currents since the vacuum magnetic field has little shear in W7-X:

$$\iota' = \iota'_{\rm ext} + \iota'_{\rm pl},\qquad \iota'_{\rm pl} \gg \iota'_{\rm ext},\qquad \iota''_{\rm pl} \gg \iota''_{\rm ext}. \qquad (25)$$

To better mimic stellarator behaviour, we split the rotational transform into an 'external' (vacuum) and an 'internal' (plasma) parts as follows:

$$\iota_{\rm ext} = 0.88 + 0.12\rho^2 \qquad \iota_{\rm pl} = 0.12\,\exp\left[-\frac{(\rho-0.5)^2}{0.02}\right]. \qquad (26)$$

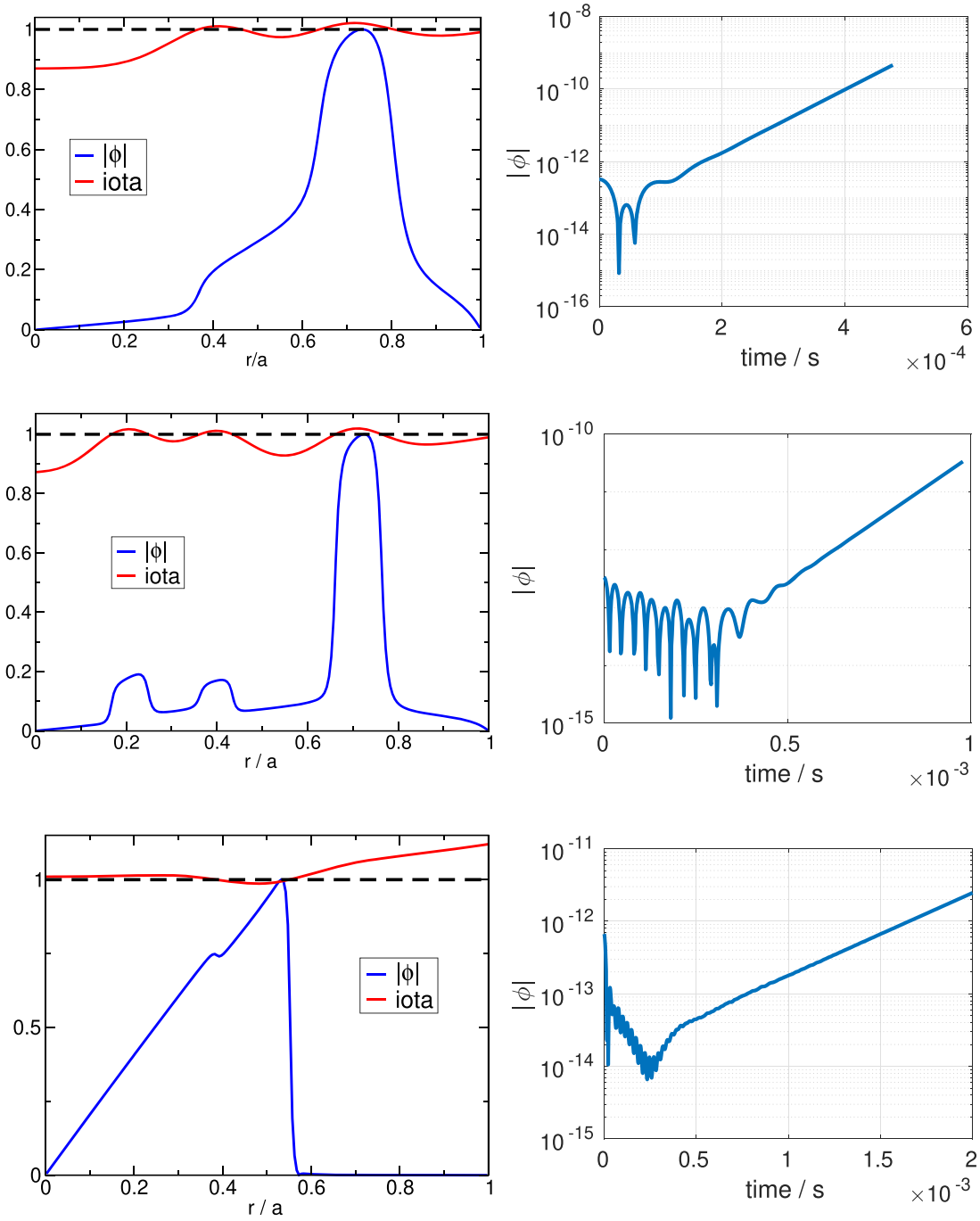


Figure 6. Mode structure (on the left) and mode evolution (on the right) for more complicated rotational transform profiles with several maxima or crossing $\iota = 1$ from above. Such profiles can be generated by the current drive in W7-X. One can see how the shear Alfvén wave continuum decays and the unstable 'multiple-kink' mode emerges in all cases. Code used: GYGLES. Model: gyrokinetics, two species, cylindrical geometry.

A similar treatment was introduced in reference [18]. When evaluating the magnetic field in equation (22), we then consider the total $\iota = \iota_{\rm ext} + \iota_{\rm pl}$, but we prescribe a plasma current, equation (23), entirely caused by $\iota_{\rm pl}$, so that

$$F(r) = \frac{B_0 r^2}{R_0}\left[\iota_{\rm ext}(r) + \iota_{\rm pl}(r)\right], \text{ but } j_{\parallel}(r) = \frac{B_0}{\mu_0 R_0}(2\iota_{\rm pl} + r\iota'_{\rm pl}). \tag{27}$$

In this way, we obtain the same resonant flux surfaces produced by the total rotational transform (27), but the plasma current is kept small, consistent with the internal part of the rotational transform. The derivatives of $\iota_{\rm pl}$ are dominant in the evaluation of the derivatives of the total current. Indeed, we see in figure 5 that the double-kink instability is driven stronger in the 'stellarator-like' case with $\gamma = 2603.1\ {\rm s}^{-1}$, when equation (27) is valid. When the total ι is used to evaluate the parallel current, equation (23), the growth rate is smaller

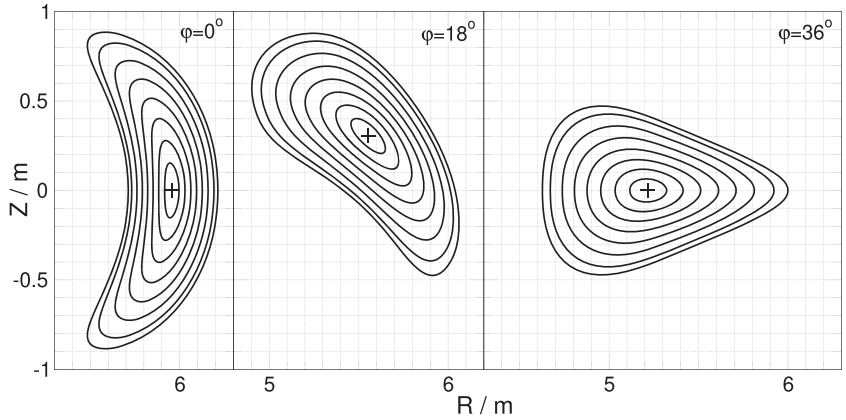


Figure 7. W7-X magnetic-flux surfaces at three characteristic cross sections, φ being the angle of the cylindrical coordinates (R, φ, Z). The magnetic axis (+) defines the $r = 0$ location, the outermost contour corresponds to $r/a = 1$, where a is the average minor radius. The magnetic equilibrium is evaluated by using the model ι and density and temperature profiles from the appendix. Code used: `VMEC`.

($\gamma = 854.5\ \mathrm{s}^{-1}$ for the case with $s_c = 0.5$ shown in figure 4), likely due to a smaller ι' in this case for $s < 0.5$ where $|\iota'_{\mathrm{pl}}|$ exceeds $|\iota'_{\mathrm{ext}} + \iota'_{\mathrm{pl}}|$. Overall, the gyrokinetic results give a milder instability, since in the ideal case we had $\gamma = 950\ \mathrm{s}^{-1}$.

In conclusion, we note that even more complicated rotational transform profiles are potentially possible for ECCD plasmas in W7-X. In figure 6, we study the model cases with two or three humps on the rotational transform profile (corresponding to four or six crossings of $\iota = 1$) as well as the case with $\iota = 1$ crossed from above (also possible in some scenarios). In all cases studied, we observe the kink instability emerging from the decaying shear Alfvén continuum with the mode structure closely following the rotational transform profile.

4.2. Hybrid fluid-kinetic treatment in real W7-X geometry

We are now ready to consider non-ideal instability in real W7-X geometry. As in the previous section, the density and temperature gradients are set to zero to eliminate their contribution to the instability drive, the study of which we leave for future work.

We initialize a series of long-wavelength modes, among which the $(m, n) = (1, -1)$ is the largest,

$$\delta\phi(\rho, \theta, \varphi) = \mathrm{e}^{\mathrm{i}(\theta - \varphi)}\, \mathrm{e}^{-\frac{1}{2}\left(\frac{\rho - 0.4}{0.1}\right)^2}, \tag{28}$$

and simulate a spectrum with $-19 \leqslant m \leqslant 11$, and $-26 \leqslant n \leqslant 24$, where m and n are the poloidal and toroidal wave-number, respectively.These values are chosen to allow no instability to grow for the highest mode numbers simulated. In all cases, the poloidal and toroidal resolution is $1/64$, whereas the radial resolution is either $1/ns = 1/256$ or $1/ns = 1/512$. In order to eliminate the pressure-gradient drive, we artificially set to zero the equilibrium density and temperature gradients, hence obtaining purely current-gradient-driven modes. Thus, for $n(0) = 2 \times 10^{19}\ \mathrm{m}^{-3}$, and $\nu_{\mathrm{ei}}/\Omega_{\mathrm{i}} = 10^{-2}$, we observe the destabilisation of a robust long-wavelength mode $(m, n) = (-4, 4)$, followed by higher mode numbers, see figure 8. Here we plot the amplitude of the radial integral of the electrostatic potential evaluated at $\varphi = 0$, (where $\varphi = 0$ defines the poloidal cut shown in the left sub-frame of figure 7) for some relevant poloidal mode numbers. An additional spectral diagnostic shows that the unstable mode $m = -4$ is indeed characterised by a toroidal mode number $n = 4$, thus resonating at the locations where $\iota = 1$. The mode belongs to the set of harmonics that dominate the solution of the global ideal MHD eigenvalue problem (for the plasma radial displacement) with the smallest eigenvalue, as evaluated in section 2, and we infer that this is a marginally stable ideal mode destabilised by non-ideal effects. The growth rate is $\gamma_{-4} = 4.276 \times 10^4\ \mathrm{s}^{-1}$. The $(m, n) = (1, -1)$ mode, whose nonlinear counterpart is experimentally observed during large sawtooth crashes in W7-X [12], is not the one most susceptible to linear instability, despite the presence of two resonant locations for which $\iota = 1$. Presumably, the $(m, n) = (1, -1)$ mode experimentally observed is the outcome of nonlinear interaction between other, linearly more unstable, modes of the same mode family. A linear analysis could not capture such type of interaction. Nevertheless, additional linear effects, such as magnetic compressibility, can also play an additional role [60].

A further comment here is in order. The most unstable eigenmode of the system generally involves many different poloidal and toroidal harmonics. In a global setting such an eigenmode may therefore be resonant ($\iota(r) = n/m$) at several radial locations and cause magnetic islands to grow on several resonant surfaces. (We remind the reader that we do not impose any particular parity on the perturbations at resonant locations, hence electromagnetic micro-instabilities can cause magnetic reconnection [61]). All our simulations are linear and, our 'most unstable mode' is simply the one with the largest growth rate. Any such an eigenmode usually involves many Fourier harmonics but may be characterised by the dominant one. In the case of figure 8, this is the $(m, n) = (-4, 4)$, but we notice that an unstable $(m, n) = (1, -1)$ mode is also present. This, as our results indicate, will eventually grow, after undergoing some oscillations, with a growth rate virtually indistinguishable from that of the $(m, n) = (-4, 4)$ mode. We are not in the position to predict whether this subtle interplay between long wavelength instabilities can be observed experimentally. We also notice moderate high-mode-number

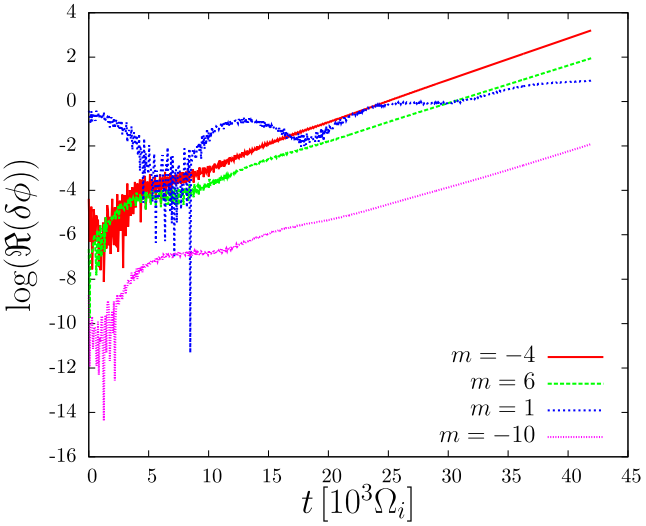


Figure 8. Most relevant modes destabilized by the equilibrium current gradient only. The complete spectrum simulated retains $-19 \leqslant m \leqslant 11$, and $-26 \leqslant n \leqslant 24$, where m and n are the poloidal and toroidal wave-number, respectively. The most susceptible to instability is the $(m, n) = (-4, 4)$. The $(m, n) = (1, -1)$ undergoes oscillations and is eventually destabilized. Moderate high-mode numbers are present. In particular one can see the $m = -10$. An alternative spectral diagnostic gives that this specific unstable mode is the $m = -10$, $n = 9$. Here $\Omega_i = 2.298 \times 10^8$ s^{-1} is the ion cyclotron frequency. Code used: EUTERPE. Model: two fluids with finite resistivity, W7-X geometry.

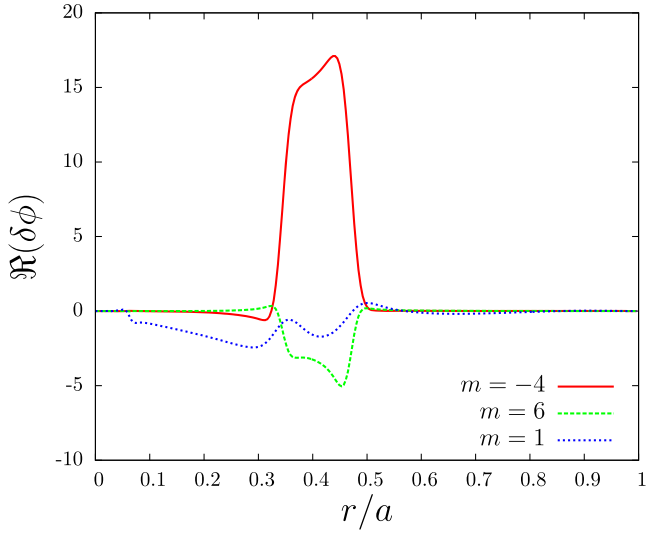


Figure 9. Radial structure of the electrostatic potential, evaluated at the specific toroidal angle $\varphi = 0$ defining the poloidal cut of the left sub-frame of figure 7, for modes with poloidal numbers $m = -4, 6, 1$ as destabilised in figure 8 for $t = 40[10^3/\Omega_i]$. Here $\Omega_i = 2.298 \times 10^8$ s^{-1} is the ion cyclotron frequency. Code used: EUTERPE. Model: two fluids with finite resistivity, W7-X geometry.

instabilities. For instance, the $m = -10$ mode grows with a growth rate $\gamma_{-10} = 3.875 \times 10^4$ s^{-1} and its dominant toroidal mode number is $n = 9$. The role of these moderate high-mode-number instabilities will be investigated in a future companion article, where the equilibrium temperature gradient will also be considered.

The long-wavelength mode shows the same qualitative structure as in the cylindrical case. In figure 9 we see the $m = -4$ component of the electrostatic potential being localized between the locations at which $\iota = 1$. The modes $m = 1$ and $m = 6$ are also plotted, for comparison. This type of modes was also found by Strumberger and Günter, for similar ι profiles, with a linear resistive MHD code [17]. To characterise the instability, we perform a scaling with plasma collisionality. Figure 10 shows such a scaling for the $m = -4$ mode in the case of flat profiles. For finite collisionality, the result follows an $\eta^{1/3}$ scaling, thus suggesting that the ECCD-perturbed ι-profile here investigated is non-ideally marginally unstable, the equivalent of the 'large Δ'-regime' in cylindrical geometry. For an infinitesimally small collisionality, an instability is obtained (for reasonable radial resolution) only if the electron mass is artificially increased. We performed a numerical test with $\nu_{ei}/\Omega_i = 10^{-6}$, setting $m_e^{NUM}/m_i = 5 \times 10^{-3}$ and rescaling the final growth rate by a factor $\sqrt{m_e/m_e^{NUM}}$. The result produces the correct collisionless asymptotic limit of γ_{-4} for $\nu_{ei} \ll 1$. At the same time, in this way, we prove that the growth rate of the current-gradient-driven instability scales as $\gamma_{-4} \sim d_e$, for $\nu_{ei} \ll 1$, [23], a result similar to the cylindrical one [58]. The hybrid version of the code is further used to study the effect of kinetic ions on this instability, which is seen to be

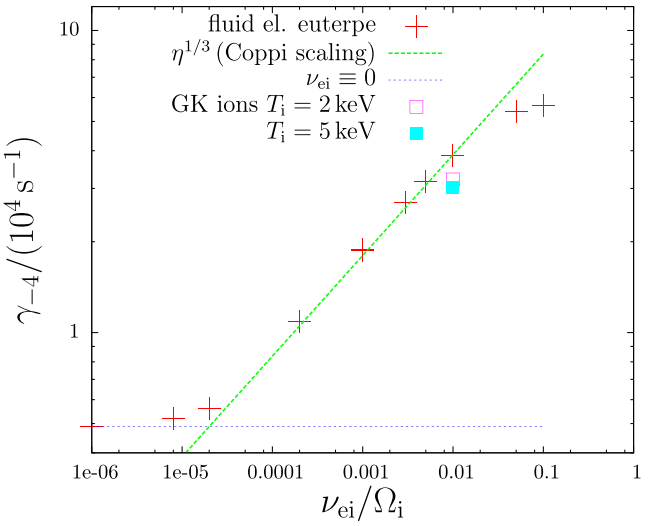


Figure 10. Scaling with collisionality, $\hat{\nu}_{ei} \equiv \nu_{ei}/\Omega_i$, for the $(m, n) = (-4, 4)$ mode for flat profiles. The Coppi scaling is recovered. W7-X is in the collisionless regimes $\hat{\nu}_{ei}(T_e \approx 5$ keV$) = 10^{-5}$, since $\nu_{ei}(T_e \approx 5$ keV$, n \approx 2 \times 10^{19}$ m$^{-3}) \approx 2.26 \times 10^3$ s^{-1}, and $\Omega_i(B = 2.5$ T$) \approx 2.4 \times 10^8$ s^{-1}. Code used: EUTERPE. Model: two fluids and hybrid with gyrokinetic ions, W7-X geometry.

stabilizing in figure 10. This is likely due to ion Landau damping setting a threshold for the destabilisation of reconnecting modes, similar to the one recently predicted by Connor *et al* [59]. This kinetic mechanism requires the reconnecting mode to have an oscillating component. While the study of fluctuations during sawtoothing operations in W7-X is possible, and has been documented [62], a detailed comparison of kinetic simulations and experiments is left for future work.

Table 1. Summary of main results for each physics model and geometry.

	Cylinder ($s_c = 0.5$)		W7-X	
Physics model	Mode numbers	Growth rates	Mode numbers	Growth rates
Ideal MHD	$(m,n) = (1,-1)$	$\gamma = 950\ \mathrm{s}^{-1}$	$(m,n) = (4,-4)$	$\gamma = 0^-\ \mathrm{s}^{-1}$
	✗	✗	$(m,n) = (4,-4)$	$\gamma \propto \eta^{1/3}$, $\nu_{ei} > \gamma$
Two fluids	✗	✗	Unstable	$\gamma \propto d_e$, $\nu_{ei} < \gamma$
Two GK species	$(m,n) = (1,-1)$	$\gamma = 854.5\ \mathrm{s}^{-1}$	✗	✗
	✗	✗	$(m,n) = (4,-4)$	$\gamma < \gamma$(two fluids)
Hybrid	✗	✗	Unstable	$\omega \neq 0$(Landau damping)

5. Discussion and perspectives

In this article we have studied current-gradient-driven long-wavelength reconnecting modes for W7-X relevant conditions, applying a hierarchy of numerical codes to describe the broad range of relevant physical properties characterising sawtoothing plasmas. On the most complete level, gyrokinetics was applied to all species to investigate the role of the rotational transform, ι, in driving magnetic reconnection. This was done with the gyrokinetic (radially) global code `GYGLES`, in cylindrical geometry. In this case, we can observe the destabilization of a 'double-kink' mode, emerging after the decay of the shear Alfvén continuum excited through the initial condition. The instability is more violent, the more plasma is involved, that is when the resonant location is increasingly far from the magnetic axis. This property derives from the MHD character of the instability, as the ideal MHD computations of section 2 show. The same qualitative mode structure is also found in real W7-X geometry. For this case we used the gyrokinetic (radially) global code `EUTERPE`, in its reduced and hybrid-fluid versions. The scaling of the long-wavelength reconnecting instability with collisionality and electron mass is the derived. It follows the typical scaling laws of the cylindrical limit, as recently derived also for stellarators [23]. More importantly, the scaling suggests that W7-X, when operating with ECCD, is driven close to ideal marginality, and therefore becomes susceptible to non-ideal instabilities. An ideal MHD analysis performed with the `CAS3D` code confirms this results. Specifically, the dominant Fourier harmonics of the ideal mode found by `CAS3D` to be closest to marginality are those first observed with the code `EUTERPE` to be non-ideally destabilised. This is useful to know, especially for future studies of pressure-driven instabilities (whether reconnecting or not). Preliminary computations suggest that the pressure-gradient drive can be quite significant in plasmas with rotational transform profiles of the type considered above. When kinetic effects are accounted for, the instability threshold can be below the ideal one. A summary of the most important results is given in table 1.

The existence of the instabilities investigated in this paper has several practical implications. Firstly, they limit the parameter space available for safe device operation, if ECCD is present, since the very largest crashes encompass the entire plasma. For a crash of this type, the diamagnetic plasma current suddenly disappears and a corresponding current in any surrounding conductors is induced. Ensuing forces could cause damage, and they should be assessed. Secondly, smaller-amplitude sawtooth oscillations cause a degradation of confinement, just like in tokamaks, by expelling the hot plasma from the plasma core. Thirdly, for future stellarators the existence of reconnecting instabilities raises the concern that a low-shear device is, in a sense, a delicate object, which can easily be affected by a small amount of an externally driven toroidal current, in particular if the latter is radially localized. However, we must point out that, in the absence of a current drive, the presence of an $\iota = 1$ surface within the plasma does not pose a limit, nor a risk, for plasma operations. This has been proven in a series of $\iota-$scan experiments in W7-X, in which the presence of rational surfaces affects, or even improves, but does not limit confinement [63]. Finally, we must point out that the understanding of the physics of crashes offers also the opportunity to control them, with all their properties. The clearest examples in which sawtooth crashes can be useful would be the removal of impurities from the core, or their exploitation for improved edge plasma fueling.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. This work has been carried out within the framework of the EUROfusion Consortium and has received funding from the Euratom research and training programme 2014–2018 and 2019–2020 under Grant Agreement No. 633053. The views and opinions expressed herein do not necessarily reflect those of the European Commission.

Appendix A. Equilibrium parameters

In sections 2 and 4.1, straight screw-pinch equilibria are studied. The basic parameters are chosen in accordance to W7-X parameters, i.e. the minor radius $a = 0.55$ m, the aspect ratio $A = L/a = 2\pi \times 10$, and the magnetic field $B_0 = 2.5$ T. We recall the model rotational transform, thus embedding (21),

$$\iota(\rho) = \iota_0 + \iota_2\rho^2 + \iota_4\rho^4 + f_{\exp}\exp\left[-\frac{(\rho - s_c)^2}{\sigma}\right], \quad \text{(A1)}$$

Table 2. Parameters of a humped rotational transform profile as function of the normalized minor radius, as given in equation (A1).

Case	Section	ι_0	ι_2	ι_4	$f_{\rm exp}$	$s_{\rm c}$	σ
Cylinder	2	0.88	0.115	0	0.2	0.4922	0.0125
		0.88	0.115	0	0.2	0.393	0.0145
		0.88	0.115	0	0.2	0.2945	0.0165
		0.88	0.115	0	0.12	0.4772	0.0357
	4.1	0.88	0.12	0	0.12	0.5	0.02
		0.88	0.12	0	0.12	0.4	0.02
		0.88	0.12	0	0.12	0.3	0.02
W7-X	2	0.86	0.042	0.072	0.1545	0.40283	0.02399
	4.1	0.86	0.042	0.072	0.1545	0.4	0.025

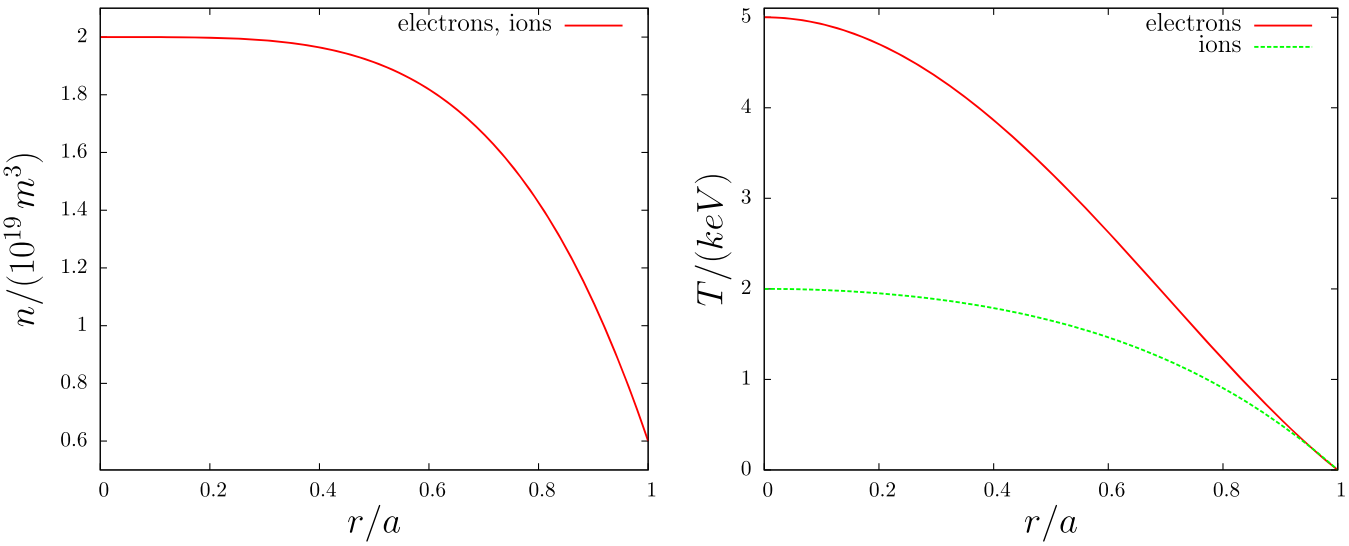


Figure 11. W7-X profiles used to evaluate the magnetic equilibrium of section 4.2: density profile (left), temperature profiles (right).

with $\rho = r/a$ the normalized minor radius, $s_{\rm c}$ the center of the ι-hump modeling ECCD, and $f_{\rm exp}$ and σ controlling its height and width. Table 2 summarizes the values of these parameters as used in sections 2 and 4.1. A flat density profile $n_0 = 2 \times 10^{19}$ m^{-3}, and flat ion and electron temperatures, $T_{\rm i} = 2$ keV and $T_{\rm e} = 5$ keV complete the equilibrium description for section 4.1. But for the flat profile shape, this choice uses plasma densities and temperatures values as measured in the recent W7-X experimental campaign [1]. Assuming quasi-neutrality, a central pressure of $p_0 = n_0(T_{\rm i} + T_{\rm e})$ is used for the ideal MHD calculations, section 2.

In sections 2 and 4.2, W7-X-type equilibria are considered using the equilibrium number density

$$n_0 = n_0(0)\left(1 - 0.7\rho^4\right), \tag{A2}$$

and the electron and ion temperatures

$$T_{\rm e} = T_{\rm e}(0)\left(1 - 1.5\rho^2 + 0.5\,\rho^4\right) \tag{A3}$$

and

$$T_{\rm i} = T_{\rm i}(0)\left(1 - 0.6\rho^2 - 0.4\,\rho^4\right), \tag{A4}$$

with $n_0(0) = 2 \times 10^{19}$ m^{-3}, $T_{\rm e}(0) = 5$ keV, $T_{\rm i}(0) = 2$ keV. These profiles are shown in figure 11. They are models designed to reproduce typical ones, see [12]. For a W7-X-type equilibrium with effective minor radius $a = 0.55$ m and aspect ratio $A = R/a = 10$, the resulting average plasma-β is $\langle\beta\rangle \approx 0.004$ with a peaking factor of $\beta(\text{axis})/\langle\beta\rangle \approx 2.46$. Unlike screw-pinch plasmas, stellarator plasmas possess rotational transform already without a net toroidal current. For a net-current-free W7-X standard case with $5/6 < \iota < 5/5$, enclosed toroidal flux $\Phi_{\rm edge} = 1.95$ V s, and the pressure profile $p_0 = n_0(T_{\rm i} + T_{\rm e})$ resulting from (A2)–(A4), the rotational transform is determined in a free-boundary `VMEC` [38, 39] calculation, giving the polynomial part of ι in (A1) with the parameters ι_0, ι_2, and ι_4 in table 2, rows referring to W7-X. Similarly to figure 3, this rotational transform profile is modified with an ι-hump, parameters $f_{\rm exp}$, $s_{\rm c}$, and σ in table 2, so as to mimic the effect of ECCD [12].

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References

[1] Wolf R.C. *et al* 2019 *Phys. Plasmas* **26** 082504
[2] Klinger T. *et al* 2017 *Plasma Phys. Control. Fusion* **59** 014018
[3] Pedersen T.S. *et al* 2019 *Plasma Phys. Control. Fusion* **61** 014035
[4] Grieger G. *et al* 1992 *Phys. Fluids* B **4** 2081
[5] Dinklage A. *et al* 2018 *Nat. Phys.* **14** 855
[6] Hirsch M. *et al* 2017 *Nucl. Fusion* **57** 086010
[7] Fuchert G. *et al* 2018 *Nucl. Fusion* **58** 106029
[8] Bozhenkov S. *et al* 2020 *Nucl. Fusion* **60** 066011
[9] Wegner T. *et al* 2020 *Nucl. Fusion* **60** 124004
[10] Nührenberg C. 1996 *Phys. Plasmas* **3** 2401
[11] Nührenberg J. and Zille R. 1988 *Phys. Lett.* A **129** 113–7
[12] Zanini M. *et al* 2020 *Nucl. Fusion* **60** 106021
[13] Wesson J. 2011 *Tokamaks* 4th edn (Oxford: Oxford University Press) ch 6.13
[14] Hastie R.J. 1997 *Astrophys. Space Sci.* **256** 177
[15] Porcelli F., Boucher D. and Rosenbluth M.N. 1996 *Plasma Phys. Control. Fusion* **38** 2163
[16] Jardin S.C., Krebs I. and Ferraro N. 2020 *Phys. Plasmas* **27** 032509
[17] Strumberger E. and Günter S. (The W7-X team) 2020 *Nucl. Fusion* **60** 106013
[18] Yu Q. *et al* 2020 *Nucl. Fusion* **60** 076024
[19] Zocco A., Mishchenko A. and Koenies A. 2019 *J. Plasma Phys.* **85** 905850607
[20] Ishida Y., Ishizawa A., Imadera K., Kishimoto Y. and Nakamura Y. 2020 *Phys. Plasmas* **27** 092302
[21] Nikulsin N., Hoelzl M., Zocco A., Lackner K. and Günter S. 2019 *Phys. Plasmas* **26** 102109
[22] Czarny O. and Huysmans G. 2008 *J. Comput. Phys.* **227** 7423
[23] Zocco A., Helander P. and Weitzner H. 2020 *Plasma Phys. Control. Fusion* **63** 025001
[24] Schep T.J., Pegoraro F. and Kuvshinov B.N. 1994 *Phys. Plasmas* **1** 2843
[25] Yu Q., Günter S. and Scott B.D. 2003 *Phys. Plasmas* **10** 797
[26] Halpern F.D., Lütjens H. and Luciani J. 2011 *Phys. Plasmas* **18** 102501
[27] Loureiro N.F., Cowley S.C., Dorland W.D., Haines M.G. and Schekochihin A.A. 2005 *Phys. Rev. Lett.* **95** 235003
[28] Loureiro N.F., Schekochihin A.A. and Zocco A. 2013 *Phys. Rev. Lett.* **111** 025002
[29] Zocco A. and Schekochihin A.A. 2011 *Phys. Plasmas* **18** 102309
[30] Mishchenko A. and Zocco A. 2012 *Phys. Plasmas* **19** 122104
[31] Fivaz M., Brunner S., de Ridder G., Sauter O., Tran T.M., Vaclavik J., Villard L. and Appert K. 1998 *Comput. Phys. Commun.* **111** 27
[32] Hatzky R., Kleiber R., Könies A., Mishchenko A., Borchardt M., Bottino A. and Sonnendrücker E. 2019 *J. Plasma Phys.* **85** 905850112
[33] Kornilov V., Kleiber R., Hatzky R., Villard L. and Jost G. 2004 *Phys. Plasmas* **11** 3196
[34] Ara G., Basu B., Coppi B., Laval G., Rosenbluth M. and Waddell B. 1978 *Ann. Phys., NY* **112** 443
[35] Schwab C. 1993 *Phys. Fluids* B **5** 3195
[36] Bernstein I.B., Frieman E.A., Kruskal M.D. and Kulsrud R.M. 1958 *Proc. R. Soc.* A **244** 17
[37] Hain K., Lüst R. and Schlüter A. 1957 *Z. Naturforsch.* A **12** 833
[38] Hirshman S.P. and Whitson J.C. 1983 *Phys. Fluids* **26** 3553
[39] Hirshman S., van RIJ W.I. and Merkel P. 1986 *Comput. Phys. Commun.* **43** 143
[40] Nührenberg C. 2016 *Nucl. Fusion* **56** 076010
[41] Bussac M.N., Pellat R., Edery D. and Soule J.L. 1975 *Phys. Rev. Lett.* **35** 1638
[42] Nührenberg C. and Boozer A.H. 2003 *Phys. Plasmas* **10** 2840
[43] Hornsby W.A, Migliano P., Buchholz R., Kroenert L., Weikl A., Peeters A.G., Zarzoso D., Poli E. and Casson F.J. 2015 *Phys. Plasmas* **22** 022118
[44] Mishchenko A., Cole M., Kleiber R. and Könies A. 2014 *Phys. Plasmas* **21** 052113
[45] Mishchenko A., Könies A., Kleiber R. and Cole M. 2014 *Phys. Plasmas* **21** 092110
[46] de Boor C. 1978 *A Practical Guide to Splines* (New York: Springer)
[47] Höllig K. 2003 *Finite Element Methods with B-Splines* (Philadelphia, PA: Society for Industrial and Applied Mathematics)
[48] Mishchenko A., Hatzky R. and Könies A. 2004 *Phys. Plasmas* **11** 5480
[49] Mishchenko A., Könies A. and Hatzky R. 2004 *Proc. Joint Varenna-Lausanne Int. Workshop* 30 August-3 Septembe Varenna, Italy (Bologna: Società Italiana di Fisica)
[50] Mishchenko A., Könies A. and Hatzky R. 2005 *Phys. Plasmas* **12** 062305
[51] Hatzky R., Könies A. and Mishchenko A. 2007 *J. Comput. Phys.* **225** 568
[52] Cole M., Mishchenko A., Konies A., Kleiber R. and Borchardt M. 2014 *Phys. Plasmas* **21** 072123
[53] Zocco A., Loureiro N.F., Dickinson D., Numata R. and Roach C.M. 2015 *Plasma Phys. Control. Fusion* **57** 065008
[54] Loureiro N., Dorland W., Fazendeiro L., Kanekar A., Mallet A., Vilelas M. and Zocco A. 2016 *Comput. Phys. Commun.* **206** 45
[55] McClenaghan J., Lin Z., Holod I., Deng W. and Wang Z. 2014 *Phys. Plasmas* **21** 122519
[56] Pegoraro F., Porcelli F. and Schep T.J. 1989 *Phys. Fluids* B **1** 364
[57] Connor J.W., Hastie R.J. and Zocco A. 2012 *Plasma Phys. Control. Fusion* **54** 035003
[58] Porcelli F. 1991 *Phys. Rev. Lett.* **66** 425
[59] Connor J.W., Ham C.J., Hastie R.J. and Zocco A. 2019 *J. Plasma Phys.* **85** 905850204
[60] Graves J.P., Zullino D., Brunetti D., Lanthaler S. and Wahlberg C. 2019 *Plasma Phys. Control. Fusion* **61** 104003
[61] Connor J.W., Hastie R.J. and Zocco A. 2013 *Plasma Phys. Control. Fusion* **55** 125003
[62] Rahbarnia K. *et al* 2020 *Plasma Phys. Control. Fusion* **63** 015005
[63] Adreeva T. 2021 (Greifswald: Institute for Plasma Physics) private communication